~~Ka~~tarzyna Soja CHE 312 HW 2 9-16-19

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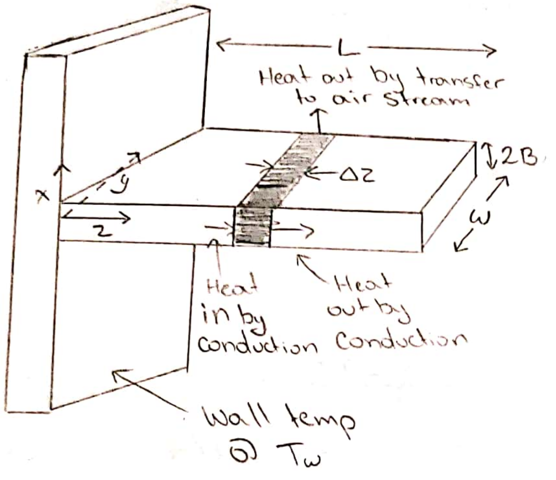
Perfect!!

Problem 1) Heat Conduction in a cooling fin

a.



The true Situation	Model We Created
1) T is a function of x, y, z but the dependence on z is most important.	1) T is a function of only z, so we ignore x and y.
2) A small amount of heat is lost at end of fin (area 2BW) and at edges (area (2BL+2BL))	2) We assume no heat is lost from the end and edges.
3) Heat transfer coefficient is a function of position.	3) Heat flux at surface is given by $q_z = h(T-T_a)$ where h=constant and $T = T(z)$.

Energy balance is made over a segment of Δz.
~~Since bar is stationary, the terms 'v' in combined energy flux 'e' may be discarded, and only contribution to energy flux~~

$$\left\{\begin{matrix}\text{Heat in}\\ \text{by}\\ \text{Conduction}\end{matrix}\right\} - \left\{\begin{matrix}\text{Heat out}\\ \text{by}\\ \text{Conduction}\end{matrix}\right\} - \left\{\begin{matrix}\text{Heat out by}\\ \text{transfer}\\ \text{to air stream}\end{matrix}\right\} = 0$$

c. $$\frac{(2BW)q_z]_z - (2BW)q_z]_{z+\Delta z}}{2BW\Delta z} - \frac{h(\cancel{2W\Delta z})(T-T_a)}{2B\cancel{W\Delta z}} = 0$$

$$\Rightarrow \frac{(\cancel{2BW})(q_z]_z - q_z]_{z+\Delta z})}{\cancel{2BW}\Delta z} = \frac{h(T-T_a)}{B}$$

$$\Rightarrow \lim_{\Delta z \to 0} \frac{(q_z]_z - q_z]_{z+\Delta z})}{\Delta z} = \frac{h(T-T_a)}{B}$$

$$\Rightarrow -\frac{dq_z}{dz} = \frac{h(T-T_a)}{B}$$

INSERT FOURIER'S LAW

$$q_z = -k\frac{dT}{dz}$$ where k = Constant

$$\Rightarrow \boxed{\frac{d^2T}{dz^2} = \frac{h(T-T_a)}{kB}}$$

Now we evaluate at following boundary conditions:

BC 1) @ $z=0$, $T=T_w$

BC 2) @ $z=L$, $\frac{dT}{dz}=0$

From here we will introduce the following dimensionless quantities:

$$\Theta = \frac{T-T_a}{T_w-T_a}$$ (Dimensionless Temp) $$\zeta = \frac{z}{L}$$ (Dimensionless Distance)

$$N^2 = \frac{hL^2}{kB}$$ (Dimensionless heat transfer coefficient)

If we substitute these terms we get:

$$\Theta(T_w - T_a) = T - T_a$$ (Take derivative of both sides wrt z)

$$\frac{d(T-T_a)}{dz} = \frac{d\Theta}{dz}(T_w - T_a)$$

$$\Rightarrow \frac{d^2T}{dz^2} = \frac{d^2\Theta}{dz^2}(T_w - T_a)$$

$$\Rightarrow \frac{h}{Bk}(T-T_a) = \frac{d^2\Theta}{dz^2}(T_w - T_a)$$

$$\Rightarrow \frac{h}{Bk}\Theta\cancel{(T_w - T_a)} = \frac{d^2\Theta}{dz^2}\cancel{(T_w - T_a)}$$

$$\Rightarrow \frac{h}{Bk}\Theta = \frac{d^2\Theta}{dz^2}$$ And we know that: $\frac{d\Theta}{d\mathcal{L}} = \frac{d\Theta}{dz} \cdot \frac{dz}{d\mathcal{L}}$

$$\Rightarrow \frac{d^2\Theta}{d\mathcal{L}^2} = \frac{d}{dz}\left(\frac{d\Theta}{dz} * \frac{dz}{d\mathcal{L}}\right)\frac{dz}{d\mathcal{L}}$$

$$= \frac{d^2\Theta}{dz^2} * \frac{dz}{d\mathcal{L}} * \frac{dz}{d\mathcal{L}} \Rightarrow \left(\frac{h}{kB}\right) * (L) * (L)$$

$$\Rightarrow \frac{d^2\Theta}{d\mathcal{L}^2} = \frac{hL^2}{kB}\Theta \Rightarrow N^2\Theta \Rightarrow \boxed{\frac{d^2\Theta}{d\mathcal{L}^2} = N^2\Theta} \quad (10.7\text{-}9)$$

Substitute back in original terms to above:

$$\Rightarrow \frac{d^2\Theta}{d\mathcal{L}^2} = \frac{hL^2}{kB} * \frac{T-T_a}{T_w - T_a} \Rightarrow \text{with } \Theta\Big]_{\mathcal{L}=0} = 1 \text{ and } \frac{d\Theta}{d\mathcal{L}}\Big]_{\mathcal{L}=1} = 0$$

Apply B.C.1 at $\ell = 0$ and $\Theta = 1$

$$\frac{dT}{dz} = \frac{d\Theta}{dz}(T_w - T_a) = \frac{1}{L}\frac{d\Theta}{d\,{}^{z}\!/\!_{L}}(T_w - T_a) \Rightarrow \frac{d\Theta}{d\ell}\,\frac{(T_w - T_a)}{L}$$

Apply B.C.2 at $\ell = 1$ and $z = 1$ we get $\frac{dT}{d\ell} = 0$

$$\frac{d^2\Theta}{d\ell^2} = N^2\Theta \Rightarrow \frac{d^2\Theta}{d\ell^2} - N^2\Theta = 0$$

From Appendix C $\Rightarrow \frac{d^2y}{dx^2} - a^2y = 0 \Rightarrow y = C_1\cosh ax + C_2\sinh ax$

OR

$y = C_3e^{+ax} + C_4e^{-ax}$

$$\Rightarrow y = C_1\cosh ax + C_2\sinh ax \Rightarrow C_1\cosh(N\ell) + C_2\sinh(N\ell)$$

When you apply the BC you get:

$$C_1 = 1$$
$$C_2 = -\tanh(N)$$

$$\Rightarrow \Theta = \cosh(N\ell) + (-\tanh(N))\sinh(N\ell)$$

$$\Rightarrow \Theta = \frac{\cosh(N)\cosh(N\ell) - \sinh(N)\sinh(N\ell)}{\cosh(N)}$$

$$\Rightarrow \boxed{\Theta = \frac{\cosh N(1-\ell)}{\cosh N}}$$

#1 The effectiveness of the fin is defined by:

$$\eta = \frac{\text{actual rate of heat loss from the fin}}{\text{rate of heat loss from an isothermal fin @ } T_w}$$

$$\eta = \frac{\int_0^W \int_0^L h(T-T_a)\,dz\,dy}{\int_0^W \int_0^L h(T_w-T_a)\,dz\,dy} = \frac{\int_0^1 \otimes \, d\mathcal{L}}{\int_0^1 d\mathcal{L}}$$

$$\Rightarrow \int_0^1 \frac{\cosh(N)\cosh(N\mathcal{L}) - \sinh(N)\sinh(N\mathcal{L})}{\cosh(N)} \, d\mathcal{L}$$

$$\Rightarrow \int_0^1 \cosh(N) - \frac{\sinh(N)}{\cosh(N)} \sinh(N\mathcal{L}) \, d\mathcal{L}$$

$$\Rightarrow \frac{\sinh(N\mathcal{L})}{N} - \frac{\sinh(N)}{N\cosh(N)} \cosh(N)\Big]_0^1$$

$$\Rightarrow -\frac{N\sinh(N)\cosh(N\mathcal{L}) - N\cosh(N)\sinh(N\mathcal{L})}{N^2\cosh(N)}\Big]_0^1$$

$$\Rightarrow -\frac{N\sinh(N-N\mathcal{L})}{N^2\cosh(N)}\Big]_0^1 \Rightarrow -\frac{\sinh(N(1-\mathcal{L}))}{N\cosh(N)}\Big]_0^1$$

$$\Rightarrow \frac{1}{\cosh(N)}\left(-\frac{1}{N}\sinh(N(1-\mathcal{L})\right)\Big]_0^1$$

$$\Rightarrow \frac{1}{\cosh(N)}\left(-\frac{1}{N}\sinh(N(1-1)\right) - \frac{1}{\cosh(N)}\left(-\frac{1}{N}\sinh(N(1-1)\right)$$

$$\Rightarrow \frac{\tanh(N)}{N} \Rightarrow \eta = \frac{\tanh(N)}{N}$$

b. Given $T_a = 350°F$

$T_w = 500°F$

$k_{FIN} = 60 \frac{Btu}{hr \cdot ft \cdot F}$

$k_{AIR} = .0022 \frac{Btu}{hr \cdot ft \cdot F}$

$h = 120 \frac{Btu}{hr \cdot ft^2 \cdot F}$

$L_{FIN} = 0.2\,ft$

$W_{FIN} = 1.0\,ft$

Thickness = 0.16 in

From Table 9.1-5 we find:

$K_{COPPER} = 379.9\ W/m \cdot K = 219.65\ Btu/hr \cdot ft \cdot F$

$K_{WOOD} = 0.126\ W/m \cdot K = 0.073\ Btu/hr \cdot ft \cdot F$

$$Q = 2WLh(T_w - T_a) \cdot \eta \quad \text{where} \quad \eta = \frac{\tanh(N)}{N}$$

$$\text{AND} \quad N = \sqrt{\frac{hL^2}{kB}}$$

$\boxed{\text{FOR COPPER}}$

$$N = \sqrt{\frac{hL^2}{kB}} = \sqrt{\frac{(120\ Btu/hr \cdot ft^2 \cdot F)(.2ft)^2}{(219.65\ Btu/hr \cdot ft \cdot F)(.08/12ft)}} = 1.8105$$

$$\eta = \frac{\tanh(1.8105)}{1.8105} = .5235$$

$$Q_{COPPER} = 2(1ft)(0.2ft)(120\ Btu/hr \cdot ft \cdot F)(500F - 350F)(.5235)$$

$$Q_{COPPER} = 3769.5\ Btu/hr$$

OR WOOD

$$N = \sqrt{\frac{hL^2}{kB}} = \sqrt{\frac{(120\ \mathrm{Btu/hr\cdot ft^2\cdot F})(.2\ \mathrm{ft})^2}{(.073\ \mathrm{Btu/hr\cdot ft\cdot F})(.08/12\ \mathrm{ft})}} = 99.312$$

$$\eta = \frac{\tanh(99.312)}{99.312} = .01006$$

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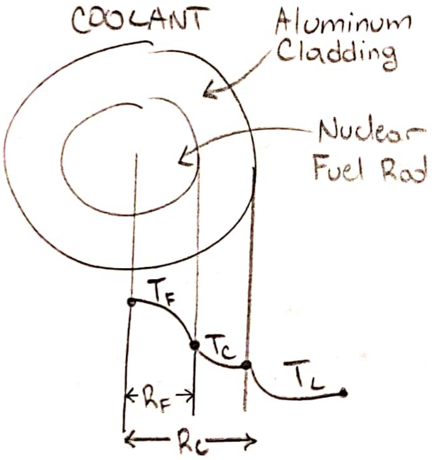
$$Q_{WOOD} = 2(1\ \mathrm{ft})(.2\ \mathrm{ft})(120\ \mathrm{Btu/hr\cdot ft^2\cdot F})(500F - 350F)(.01006)$$

$$Q_{WOOD} = 72.5\ \mathrm{Btu/hr}$$

Problem 2) Solve 10B.1

a.



Assumptions:

- Suspended in large motionless fluid
- k is constant
- Steady State

Energy Balance:

$$\left\{\begin{matrix}\text{RATE OF}\\ \text{ENERGY}\\ \text{IN}\end{matrix}\right\} - \left\{\begin{matrix}\text{RATE OF}\\ \text{ENERGY}\\ \text{OUT}\end{matrix}\right\} + \cancelto{0}{\{\text{GEN.}\}} - \cancelto{0}{\{\text{CON.}\}} = \cancelto{0}{\{\text{ACCUM.}\}}$$

$$\Rightarrow 4\pi r^2 q_r\Big]_r - 4\pi r^2 q_r\Big]_{r+\Delta r} = 0 \quad \text{(Divide by } 4\pi\Delta r\text{)}$$

$$\Rightarrow \frac{r^2 q_r\Big]_r - r^2 q_r\Big]_{r+\Delta r}}{\Delta r} = 0 \quad \text{(Take Limit)}$$

$$\Rightarrow \lim_{\Delta r \to 0} \frac{r^2 q_r\Big]_r - r^2 q_r\Big]_{r+\Delta r}}{\Delta r} = 0$$

We get the following differential:

CONSTANT IN PROBLE

$$\frac{d}{dr}(r^2 q_r) = 0 \quad \text{and we know that} \quad q_r = -k\frac{dT}{dr}$$

We can substitute Fourier's Law into our differential:

$$\frac{d}{dr}\left(r^2 \frac{dT}{dr}\right) = 0$$

b. $\int r^2 \frac{dT}{dr} = \int 0 \cdot dr \Rightarrow \frac{r^2 \frac{dT}{dr}}{r^2} = \frac{C_1}{r^2} \Rightarrow \int dT = \int \frac{C_1}{r^2} dr \Rightarrow T = \frac{-C_1}{r} + C_2$

BC 1) $r = R$, $T = T_R$

BC 2) $r \to \infty$, $T = T_\infty$

If we apply BC 2 first, we know C_1 will go to zero leaving

$T_\infty = C_2$

Apply BC 1 $\Rightarrow T_R = \frac{-C_1}{R} + T_\infty \Rightarrow T_R - T_\infty = \frac{-C_1}{R} \Rightarrow C_1 = -R(T_R - T_\infty)$

Plug everything back in:

$$T = \frac{+R(T_R - T_\infty)}{r} + T_\infty \Rightarrow T - T_\infty = \frac{R(T_R - T_\infty)}{r} \Rightarrow \frac{R}{r} = \frac{T - T_\infty}{T_R - T_\infty}$$

c. Nusselt Number $\Rightarrow Nu = \frac{hD}{k} = 2$

$q_r\Big]_{r=R} = -k \frac{dT}{dr}\Big]_{r=R}$ Where $\frac{dT}{dr} = \frac{-R(T_R - T_\infty)}{r^2}$

$$\Rightarrow q_r\Big]_{r=R} = \frac{+kR(T_R - T_\infty)}{r^2}\Bigg]_{r=R} = \frac{k(T_R - T_\infty)}{R}$$ (Equate to Newtons Law of Cooling)

$$\Rightarrow \frac{k(T_R - T_\infty)}{R} = h(T_R - T_\infty) \Rightarrow 1 = \frac{hR}{k}$$ Where $R = D/2$

$\# \Rightarrow 1 = \dfrac{hD}{2k} \Rightarrow 2 = \dfrac{hD}{k} = Nu$

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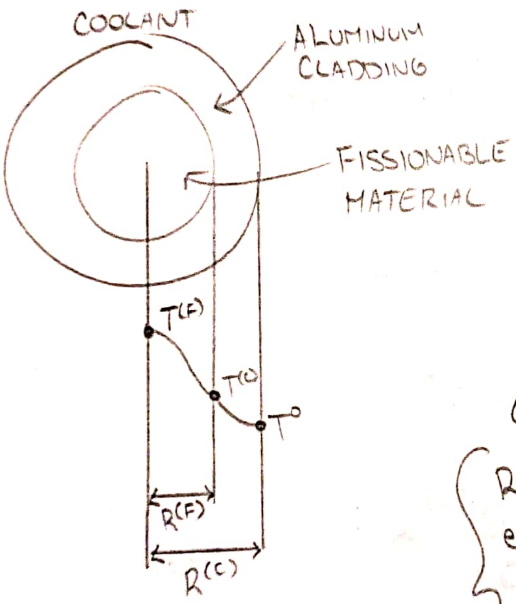
d. For Nusselt number the k in the denominator is only for liquids while the Biot number has a k for solids. They have similar ~~formulas~~ but calculate different things.

Problem 3)



$$S_N = \frac{S_{N0}}{1-\beta^2}\left[1+\beta\left(\frac{r}{R_F}\right)^3\right]$$

Where S_{N0} and β = CONSTANT

Assumptions:

- Steady State
- Sphere not in motion

ENERGY BALANCE:

$$\begin{Bmatrix}\text{Rate of}\\\text{energy}\\\text{in}\end{Bmatrix} - \begin{Bmatrix}\text{Rate of}\\\text{energy}\\\text{out}\end{Bmatrix} + \begin{Bmatrix}\text{Rate of}\\\text{thermal}\\\text{energy}\end{Bmatrix} = 0$$

$$\Rightarrow \frac{\cancel{4\pi} r^2 q_r^{(F)}\Big]_r - \cancel{4\pi} r^2 q_r^{(F)}\Big]_{r+\Delta r}}{\cancel{4\pi}\Delta r} + \frac{S_n \cancel{4\pi} r^2 \cancel{\Delta r}}{\cancel{4\pi \Delta r}} = 0$$

$$\Rightarrow \lim_{\Delta r \to 0} \frac{r^2 q_r^{(F)}\Big]_{r+\Delta r} - r^2 q_r^{(F)}\Big]_r}{\Delta r} = S_n r^2$$

$$\Rightarrow \frac{d}{dr}\left(r^2 q_r^{(F)}\right) = \left[\frac{S_{n0}}{1-\beta^2}\left[1+\beta\left(\frac{r}{R_F}\right)^3\right]\right] r^2$$

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$$\Rightarrow \int d\left(r^2 q_r^{(F)}\right) = \int \left(\frac{S_{no}}{1-\beta^2}\left[r^2 + \beta\left(\frac{r^5}{R_F^3}\right)\right]\right) dr$$

$$\Rightarrow r^2 q_r^{(F)} = \frac{S_{no}}{1-\beta^2}\left[\frac{r^3}{3} + \beta\left(\frac{r^6}{6R_F^3}\right)\right] + C_1^{(F)}$$ (Divide out r^2)

$$\Rightarrow q_r^{(F)} = \frac{S_{no}}{1-\beta^2}\left[\frac{r}{3} + \beta\left(\frac{r^4}{6R_F^3}\right)\right] + \frac{C_1^{(F)}}{r^2}$$

We can show the same work for $q_r^{(c)}$ but there is no significant source term:

$$\frac{d}{dr}\left(r^2 q_r^{(c)}\right) = 0 \Rightarrow \int d\left(r^2 q_r^{(c)}\right) = \int 0\, dr$$

$$\Rightarrow r^2 q_r^{(c)} = C_1 \Rightarrow q_r^{(c)} = \frac{C_1^{(c)}}{r^2}$$

Now we can plug in our Boundary Conditions:

BC1) When $r=0$, $q_r^{(F)}$ is not infinate

BC2) When $r=R^{(F)}$, $q_r^{(F)} = q_r^{(c)}$

For BC1) we can not divide by zero so $C_1^{(F)} = 0$, leaving us w

$$q_r^{(F)} = \frac{S_{no}}{1-\beta^2}\left[\frac{r}{3} + \beta\left(\frac{r^4}{6R_F^3}\right)\right]$$

For BC2) $\Rightarrow \frac{S_{no}}{1-\beta^2}\left[\frac{r}{3} + \beta\left(\frac{r^4}{6R_F^3}\right)\right] = \frac{C_1^{(c)}}{r^2}$ (Plug in $r = R^{(F)}$)

$$\Rightarrow \frac{S_{no}}{1-\beta^2}\left[\frac{R^{(F)}}{3} + \beta\left(\frac{R^{(F)4}}{6R^{(F)3}}\right)\right] = \frac{C_1^{(c)}}{R^{(F)2}}$$

$$\Rightarrow C_1^{(c)} = \frac{S_{no}}{1-\beta^2}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right]$$

$$\Rightarrow q_r^{(c)} = \frac{\frac{S_{no}}{1-\beta^2}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right]}{r^2}$$

Plug in Fourier's Law into $q_r^{(c)}$ and $q_r^{(F)}$:

$$q_r^{(F)} = -k^{(F)}\frac{dT^{(F)}}{dr} \Rightarrow -k^{(F)}\frac{dT^{(F)}}{dr} = \frac{S_{no}}{1-\beta^2}\left[\frac{r}{3} + \beta\left(\frac{r^4}{6R_F^3}\right)\right]$$

$$\Rightarrow \int dT^{(F)} = \int -\frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{r}{3} + \beta\left(\frac{r^4}{6R_F^3}\right)\right] dr$$

$$\Rightarrow T^{(F)} = -\frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{r^2}{6} + \beta\left(\frac{r^5}{30R_F^3}\right)\right] + C_2^{(F)}$$

$$q_r^{(c)} = -k^{(c)}\frac{dT^{(c)}}{dr} \Rightarrow -k^{(c)}\frac{dT^{(c)}}{dr} = \frac{\frac{S_{no}}{1-\beta^2}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right]}{r^2}$$

$$\Rightarrow \int dT^{(c)} = \int -\frac{S_{no}}{k^{(c)}(1-\beta^2)\, r^2}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right] dr$$

$$T^{(c)} = \frac{S_{no}}{k^{(c)}r(1-\beta^2)}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right] + C_2^{(c)}$$

Now we plug in new Boundary Conditions:

BC 3) when $r = R^{(F)}$, $T^{(F)} = T^{(c)} = T^{(c)}_{MAX}$

BC 4) when $r = R^{(c)}$, $T^{(c)} = T_0$

For BC 3) $\Rightarrow$ $C_2^{(c)} = T_o - \dfrac{S_{no}}{k^{(c)}R^{(c)}(1-\beta^2)}\left[\dfrac{R^{(F)3}}{3} + \beta\dfrac{R^{(F)3}}{6}\right]$

$$T^{(c)} = \frac{S_{no}}{k^{(c)}r(1-\beta^2)}\left[\frac{R^{(F)3}}{3} + \beta\left(\frac{R^{(F)3}}{6}\right)\right] + T_o - \frac{S_{no}}{k^{(c)}R^{(c)}(1-\beta^2)}\left[\frac{R^{(F)3}}{3} + \beta\frac{R^{(F)3}}{6}\right]$$

$$T^{(C)} = T^{(C)}_{MAX} = \frac{S_{no}R^{(F)3}}{k^{(c)}(1-\beta^2)}\left(\frac{1}{3} + \beta\left(\frac{1}{6}\right)\right)\left[\frac{1}{r} - \frac{1}{R^{(c)}}\right] + T_o$$

For BC 4) $\Rightarrow$

$$T_o + \frac{S_{no}R^{(F)3}}{k^{(c)}(1-\beta^2)}\left(\frac{1}{3} + \beta\left(\frac{1}{6}\right)\right)\left[\frac{1}{R^{(c)}} - \frac{1}{R^{(c)}}\right] = -\frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{r^2}{6} + \beta\left(\frac{r^5}{30R_F^3}\right)\right] + C_2^{(F)}$$

$$C_2^{(F)} = T_o + \frac{S_{no}R^{(F)3}}{k^{(c)}(1-\beta^2)}\left(\frac{1}{3} + \beta\left(\frac{1}{6}\right)\right) + \frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{R^{(c)}}{6} + \beta\left(\frac{R^{(c)5}}{30R_F^3}\right)\right]$$

$$T^{(F)} = T_o + \frac{S_{no}R^{(F)3}}{k^{(c)}(1-\beta^2)}\left(\frac{1}{3} + \beta\left(\frac{1}{6}\right)\right) + \frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{R^{(c)}}{6} + \beta\left(\frac{R^{(c)5}}{30R_F^3}\right)\right] - \frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{r^6}{6} + \beta\frac{r^5}{30R_F^3}\right]$$

To find $T^{(F)}_{MAX}$, we know it will be where $r=0$:

$$T^{(F)}_{MAX} = T_o + \frac{S_{no}R^{(F)3}}{k^{(c)}(1-\beta^2)}\left(\frac{1}{3} + \beta\left(\frac{1}{6}\right)\right) + \frac{S_{no}}{k^{(F)}(1-\beta^2)}\left[\frac{R^{(c)}}{6} + \beta\left(\frac{R^{(c)5}}{30R_F^3}\right)\right]$$

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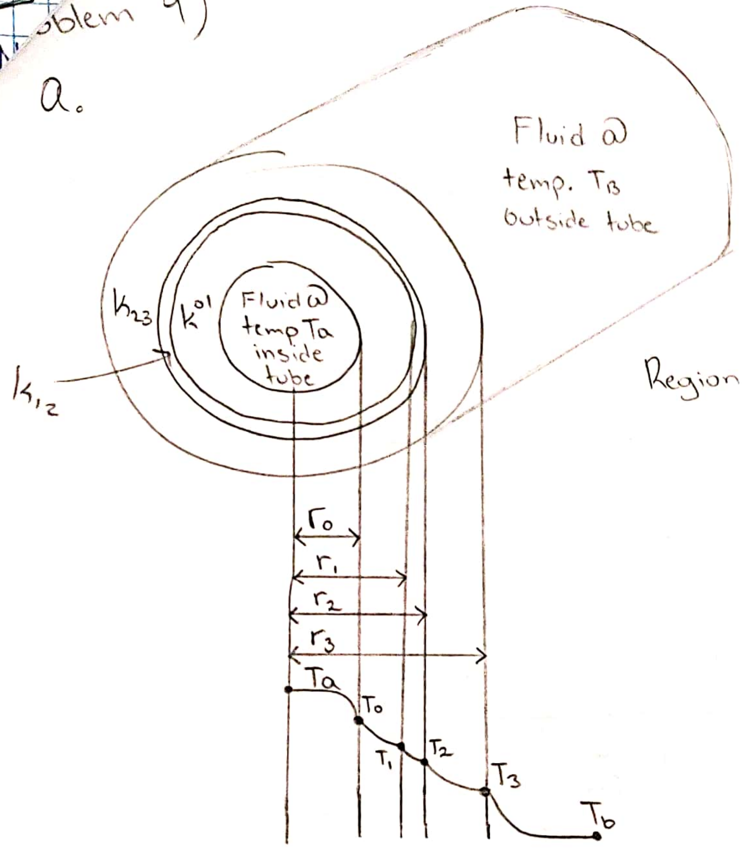
Problem 4)

a.



Assumptions:

- Steady State
- k = constant

Energy Balance:

Region 1)
$$\frac{2\pi r L q_r\big]_r - 2\pi r L q_r\big]_{r+\Delta r}}{2\pi r \Delta r L} = 0$$

$$\Rightarrow \lim_{\Delta r \to 0} \frac{r q_r\big]_r - r q_r\big]_{r+\Delta r}}{\Delta r} = 0$$

$$\Rightarrow \frac{d}{dr}(r q_r) = 0$$ First order diff. equ. for flux distribution

$$\Rightarrow \int d(r q_r) = \int 0 \cdot dr$$

$$\Rightarrow r q_r = C_1$$ $\Rightarrow$ BC 1) when $r = r_0$, $q_r = q_0$

$$\Rightarrow r q_r = r_0 q_0$$

We must apply Fourier's Law for Region 1, 2, 3 now:

Region 1) $$-k_{01} r \frac{dT}{dr} = r_0 q_0 \Rightarrow \int_1^0 dT = \int_0^1 -\frac{r_0 q_0}{r k_{01}} dr$$

Region 2) $$-k_{12} r \frac{dT}{dr} = r_0 q_0 \Rightarrow \int_2^1 dT = \int_1^2 -\frac{r_0 q_0}{r k_{12}} dr$$

Region 3) $$-k_{23} r \frac{dT}{dr} = r_0 q_0 \Rightarrow \int_3^2 dT = \int_2^3 -\frac{r_0 q_0}{r k_{23}} dr$$

Region 1) $$T_0 - T_1 = \frac{r_0 q_0}{k_{01}} \ln\left(\frac{r_1}{r_0}\right)$$

Region 2) $$T_1 - T_2 = \frac{r_0 q_0}{k_{12}} \ln\left(\frac{r_2}{r_1}\right)$$

Region 3) $$T_2 - T_3 = \frac{r_0 q_0}{k_{23}} \ln\left(\frac{r_3}{r_2}\right)$$

BC 2) $T_a - T_o = \frac{q_o}{h_o}$

BC 3) $T_3 - T_b = \frac{q_3}{h_3} = \frac{q_o}{h_3}\frac{r_o}{r_3}$

Add all last 5 equations together:

$$T_a - T_b = r_o q_o \left(\frac{1}{r_o h_o} + \frac{\ln(r_1/r_o)}{k_{01}} + \frac{\ln(r_2/r_1)}{k_{12}} + \frac{\ln(r_3/r_2)}{k_{23}} + \frac{1}{r_3 h_3} \right)$$

$$r_o q_o = \frac{T_a - T_b}{\left(\frac{1}{r_o h_o} + \sum_{j=1}^{3} \frac{\ln(r_j/r_{j-1})}{k_{j-1,j}} + \frac{1}{r_3 h_3} \right)}$$

Where $Q_o = 2\pi L r_o q_o = \frac{2\pi L (T_a - T_b)}{\left(\frac{1}{r_o h_o} + \sum_{j=1}^{3} \frac{\ln(r_j/r_{j-1})}{k_{j-1,j}} + \frac{1}{r_3 h_3} \right)}$

And we can define an "overall heat transfer coefficient based on the inner surface" U_o by:

$$Q_o = 2\pi L r_o q_o = U_o (2\pi L r_o)(T_a - T_b)$$

Combine Last 2 equations gives us:

$$\frac{1}{r_o U_o} = \left(\frac{1}{r_o h_o} + \sum_{j=1}^{3} \frac{\ln(r_j/r_{j-1})}{k_{j-1,j}} + \frac{1}{r_3 h_3} \right)$$

$$\frac{1}{r_o U_o} = \frac{1}{r_o h_o} + \frac{\ln(r_1/r_o)}{k_{01}} + \frac{\ln(r_2/r_1)}{k_{12}} + \frac{\ln(r_3/r_2)}{k_{23}} + \frac{1}{r_3 h_3}$$

Regions 01, 12, 23 $= .25\,\cancel{in} * \frac{1\,ft}{12\,\cancel{in}} = .0208\,ft$

$L = 1\,\cancel{m} * \frac{3.28\,ft}{1\,\cancel{m}} = 3.28\,ft$

$r_o = 1\,cm * \frac{1\,ft}{30.48\,cm} = .0328\,ft$

$h_o = 700\ Btu/hr \cdot ft^2 \cdot F$ $\quad T_o = 950°F$

$h_3 = 7\ Btu/hr \cdot ft^2 \cdot F$ $\quad T_3 = 50°F$

Values from Internet:

$k_{01} = 231\ Btu/hr\,ft \cdot F$

$k_{12} = 26.1\ Btu/hr \cdot ft \cdot F$

$k_{23} = 0.9\ Btu/hr \cdot ft \cdot F$

$r_1 = .0328 + .0208 = .0536$

$r_2 = .0536 + .0208 = .0744$

$r_3 = .0744 + .0208 = .0952$

$$\frac{1}{r_o U_o} = \frac{1}{(.0328ft)(700\,Btu/hr \cdot ft^2 \cdot F)} + \frac{\ln\left(\frac{.0536}{.0328}\right)}{231\,Btu/hr \cdot ft \cdot F} + \frac{\ln\left(\frac{.0744}{.0536}\right)}{26.1\,Btu/hr \cdot ft \cdot F} + \frac{\ln\left(\frac{.0952}{.0744}\right)}{.9\,Btu/hr \cdot ft \cdot F} + \frac{1}{(.0952ft)(7\,Btu/hr \cdot ft^2 \cdot F)}$$

$$\frac{1}{r_o U_o} \Rightarrow 1.832736 \frac{hr \cdot ft \cdot F}{Btu} \Rightarrow r_o U_o = .5456 \frac{Btu}{hr \cdot ft \cdot F}$$

$Q_o = 2\pi L (r_o U_o)(T_a - T_b)$

$Q_o = 2\pi(3.28\,\cancel{ft})(.5456 \frac{Btu}{hr \cdot \cancel{ft} \cdot \cancel{F}})(950F - 50\cancel{F})$

$Q_o = 10119.8\ Btu/hr$

C. We can use heat capacity equation: $q = C * m * \Delta T$

$$\Delta T = \frac{q}{C * m} \Rightarrow \frac{10119.8 \, Btu/hr}{(1.001 \, \frac{Btu}{lbm \cdot F})(10 \, lb)}$$

$\Delta T = 1010.97 \, F/hr$

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